

Physics-Guided Multi-Branch Deep Learning for Nowcasting and Short-Term Anticipation of Solar Flares Using Combined Soft and Hard X-Ray Data from Aditya-L1

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Abstract

Reliable short-term anticipation of solar flares remains an open problem in operational space weather, where the dominant approach in the literature relies on photospheric magnetogram data from missions such as the Solar Dynamics Observatory (SDO) rather than disk-integrated X-ray flux. The Soft X-ray Spectrometer (SoLEXS) and High Energy L1 Orbiting X-ray Spectrometer (HEL1OS) instruments aboard India's Aditya-L1 mission provide, for the first time from the L1 vantage point, simultaneous thermal and non-thermal X-ray measurements at cadences suited to detecting the earliest stages of the impulsive phase. This study synthesizes the current state of the art in AI-driven flare forecasting, identifies a research gap concerning X-ray-only nowcasting frameworks, and proposes a Physics-Guided Multi-Branch Long Short-Term Memory (PG-MBL) architecture that processes SoLEXS and HEL1OS streams in parallel branches while constraining the joint prediction with the Neupert effect as a physically motivated regularization term. We describe the proposed system architecture, the engineered features derived from thermal and non-thermal flux, a training strategy suited to the severe class imbalance inherent to flare catalogs, and an evaluation protocol built around the True Skill Statistic, Heidke Skill Score, False Alarm Ratio, Precision, Recall, and forecast lead time. No experimental results are reported in this study, as the described architecture has not yet been trained or validated on flight data; the contribution is a methodological framework and an accompanying experimental protocol intended to guide subsequent implementation and validation efforts using Aditya-L1 telemetry. This work is intended as a research proposal and methodological framework for future implementation and experimental validation.

Index Terms: *Aditya-L1, solar flare forecasting, SoLEXS, HEL1OS, Neupert effect, physics-guided deep learning, long short-term memory, space weather nowcasting.*

I. INTRODUCTION

Solar flares are sudden releases of magnetic energy in the solar atmosphere that produce electromagnetic radiation across the spectrum, from radio wavelengths to gamma rays, on timescales of seconds to hours. Their downstream effects include ionospheric disturbances that degrade high-frequency radio communication and GPS positioning, elevated radiation exposure for astronauts and high-altitude aircrew, and, for the most energetic X-class events, disruption of satellite electronics and terrestrial power infrastructure. The operational consequence is that space weather forecasting agencies require not only accurate classification of flare magnitude but also sufficient lead time to issue actionable warnings before the impulsive phase begins.

India's Aditya-L1 mission, positioned at the Sun–Earth L1 Lagrange point, carries two complementary X-ray payloads: the Soft X-ray Spectrometer (SoLEXS), sensitive in the 1–30 keV thermal regime, and the High Energy L1 Orbiting X-ray Spectrometer (HEL1OS), sensitive in the 10–150 keV non-thermal regime

[7]. This pairing is distinct from the single, broadband X-ray photometry historically available from missions such as GOES, and offers an opportunity to separate the thermal plasma response from the non-thermal particle acceleration that characterizes the onset of a flare. The central question this study addresses is whether, and how, this thermal/non-thermal separation can be exploited by a data-driven forecasting system to extend the effective warning window beyond what disk-integrated X-ray photometry alone has previously supported.

This study does not report new experimental results. Its contribution is a methodological synthesis: a critical review of the state of the art in AI-based flare forecasting, an explicit statement of the gap that X-ray-only nowcasting occupies relative to the magnetogram-centric literature, and a physics-guided multi-branch architecture designed specifically for the SoLEXS/HEL1OS data pair, together with a training and evaluation protocol suitable for the severe class imbalance that characterizes flare catalogs. Where the supporting literature is inferred rather than directly verified, this is stated explicitly rather than presented as established fact.

II. RELATED WORK

The transition from purely physics-based forecasting to data-driven forecasting has been the dominant trend in recent space weather literature. Magnetohydrodynamic (MHD) simulations, while grounded in first-principles physics, are computationally expensive and difficult to run at the cadence required for operational nowcasting. This has motivated a shift toward machine learning and deep learning models trained on historical observational catalogs, a trend consistent with the general movement across heliophysics toward surrogate and data-driven models [1].

A substantial portion of the existing literature is built around photospheric vector magnetogram data, most notably the Space-weather HMI Active Region Patches (SHARP) parameters derived from the Helioseismic and Magnetic Imager (HMI) aboard the Solar Dynamics Observatory. The physical rationale for this emphasis is that magnetic reconnection in the corona is the underlying energy-release mechanism for a flare, and quantities such as magnetic shear, current helicity, and the gradient of the horizontal field capture the build-up of non-potential magnetic energy that precedes an eruption, sometimes by several hours. Reported forecasting horizons in this line of work are commonly on the order of two to three hours, which reflects the timescale over which measurable changes in active-region magnetic complexity accumulate before an eruption.

A second, smaller body of work uses X-ray flux time series directly, typically from GOES, as either the sole predictor or as an auxiliary signal alongside magnetograms. Convolutional neural networks and recurrent architectures, particularly Long Short-Term Memory (LSTM) networks, have been reported to outperform classical empirical and linear-regression baselines when applied to sequential flux data, consistent with the general observation that recurrent architectures are well suited to time-dependent physical processes [2][3]. At the same time, tree-based ensemble methods such as Random Forest and XGBoost have been reported to perform competitively, and in some studies preferentially, on structured tabular features such as flare history counts, active-region area, and integrated flux over fixed windows [4]. This is not a contradiction so much as a reflection of the underlying data structure: tree-based models are well matched to engineered, tabular feature sets, while deep sequential architectures are better matched to raw, high-dimensional, or high-cadence time series where the relevant temporal structure is not easily hand-engineered.

A recurring theme across this literature is the argument that hybrid models architectures that combine a data-driven core with an explicit physical constraint, represent the most promising path toward models that are both accurate and trustworthy enough for operational deployment [1]. Physics-informed neural networks, in which a physical relationship is expressed as an additional term in the training loss rather than as a hard constraint on the architecture, have been proposed as a general strategy for encoding domain knowledge into otherwise data-driven models without discarding the flexibility that makes deep learning effective in the first place.

Within X-ray-specific studies, two physical relationships recur as candidate precursor signatures. The first is a slow, sustained rise in soft X-ray flux at a comparatively stable temperature prior to the impulsive phase, sometimes referred to descriptively as a hot-onset precursor. The second is the Neupert effect, an empirical relationship in which the time-integrated hard X-ray flux tracks the rise of the soft X-ray flux, reflecting the chromospheric evaporation process by which non-thermal electron beams heat the lower atmosphere and drive the thermal plasma response measured in soft X-rays [6]. Deviations from the expected Neupert relationship have been interpreted in the literature as indicative of differing heating mechanisms and have been suggested, though not conclusively established across a large sample, as informative for anticipating flare magnitude [6].

III. RESEARCH GAP

Three gaps in the existing literature motivate the methodology proposed in this study. First, the dominant magnetogram-centric approach is not directly transferable to the Aditya-L1 X-ray payload, which is disk-integrated and carries no information about the spatial location or magnetic topology of an active region. A forecasting framework built for SoLEXS and HELIOS must therefore be designed around what disk-integrated thermal and non-thermal flux can support, rather than adapted from an architecture designed for spatially resolved magnetic field data.

Second, prior X-ray-based studies have generally treated soft and hard X-ray channels as a single combined flux signal, following the precedent set by single-band GOES photometry. Aditya-L1 is, to the best of our knowledge based on the surveyed literature, the first mission to provide simultaneous, independently calibrated thermal (SoLEXS) and non-thermal (HELIOS) X-ray measurements from the L1 vantage point at cadences suited to flare studies. Whether explicitly separating these two energy regimes into distinct model branches yields a measurable improvement over a combined-flux baseline is, at present, an open empirical question that the surveyed literature does not directly answer, since no prior study has had access to this specific instrument pairing.

Third, while the Neupert effect is well established as a qualitative diagnostic in solar physics, its use as a quantitative constraint inside a neural network loss function for flare forecasting, as opposed to post-hoc analysis of already-observed flares is not, to our knowledge, extensively documented in the surveyed literature. This represents a methodological gap between an established physical relationship and its formal integration into a predictive, rather than purely descriptive, machine learning framework.

IV. PROBLEM STATEMENT

Given a rolling window of disk-integrated soft X-ray flux from SoLEXS and hard X-ray flux from HELIOS, the objective is to forecast, on a nowcasting-to-short-term-anticipation horizon of minutes to

approximately one hour, (i) the probability that a flare of at least a specified GOES-equivalent class will occur within a defined forward window, and (ii) where feasible, the anticipated onset time of the impulsive phase. The problem is characterized by three properties that materially constrain the choice of methodology: strong temporal dependence in the precursor signal, severe class imbalance between quiescent intervals and, in particular, M- and X-class events, and the absence of spatial information that would allow attribution of a forecast to a specific active region. The methodology proposed in Section V is designed explicitly around these three constraints rather than adapted post hoc from magnetogram-based approaches.

V. METHODOLOGY

The methodology described in this section is a proposed design, not a validated implementation. It is presented at the level of architecture, feature definitions, and training protocol so that it can be directly implemented against Aditya-L1 telemetry once sufficient labelled flare intervals are available; no claim is made here about its eventual accuracy.

A. System Architecture

This work proposes a Physics-Guided Multi-Branch LSTM (PG-MBL) with two input branches feeding a shared fusion and prediction stage.

- Branch A (thermal, SoLEXS): a stacked LSTM or BiLSTM encoder over the soft X-ray time series (1–30 keV), intended to capture the slow-rise precursor pattern and the plasma heating trend that precedes the impulsive phase.
- Branch B (non-thermal, HELIOS): a parallel LSTM or temporal-convolutional encoder over the hard X-ray time series (10–150 keV), intended to capture short-duration bursts and quasi-periodic pulsations associated with early particle acceleration.
- Fusion layer: the two branch encodings are concatenated and passed through a dense layer that produces the flare-probability output and, optionally, an auxiliary onset-time regression output.
- Physics-guided loss term: alongside the standard classification loss, a Neupert-consistency penalty is computed by comparing the time-integrated HXR flux predicted or observed within the window to the derivative of the SXR flux over the same window, discouraging the model from learning correlations that violate this established relationship [6].

The rationale for a two-branch rather than a single-input design is that SoLEXS and HELIOS are not merely two channels of the same physical quantity but measure distinct populations: thermal plasma versus non-thermal accelerated particles, governed by different, if coupled, physical processes. A single concatenated input risks allowing the model to learn spurious cross-channel correlations that do not reflect this physical coupling, whereas separate encoders followed by an explicit, physically motivated fusion constraint make the coupling assumption an explicit part of the architecture rather than an implicit hope.

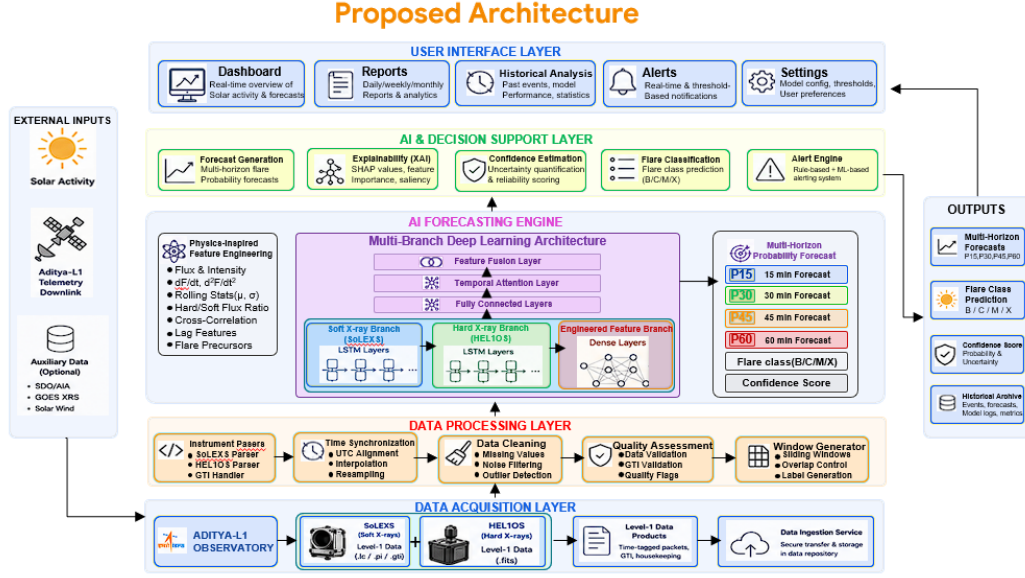


Figure 1.
Proposed Physics-Guided Multi-Branch Architecture.

B. Feature Engineering

In addition to raw flux time series, the proposed methodology introduces the following engineered features, each with an explicit physical justification:

- SXR time derivative (dS/dt): a discretized rate of change of the soft X-ray flux, serving as a proxy for the plasma heating rate and forming one side of the Neupert-consistency comparison [6].
- HXR/SXR hardness ratio: the ratio of hard-to-soft X-ray flux, intended to track the transition from thermally dominated to non-thermally dominated emission as a flare progresses through its precursor and impulsive phases.
- Emission measure (EM) and temperature (T), derived from the SoLEXS spectrum: tracked jointly to identify the characteristic movement in EM–T space, sometimes described in the literature as a horizontal-branch pattern, in which emission measure rises while temperature remains comparatively stable, a pattern reported as a pre-flare indicator in prior GOES-based studies, though its behavior in Aditya-L1 data has not yet been characterized.
- Flare-history features: counts and recency of prior M- and X-class events within a trailing window, included on the basis that active regions with recent significant flaring activity are statistically more likely to flare again, a relationship well documented in flare-catalog studies though one that, by construction, provides no information about which currently active region is responsible for the disk-integrated signal.

We note explicitly that because SoLEXS and HEL10S are disk-integrated instruments, none of the above features can be attributed to a specific active region without an auxiliary spatial data source. This limitation is discussed further in Section VIII.

C. Model Selection

Table I summarizes the trade-offs among the model families surveyed in Section II, evaluated specifically against the properties of the SoLEXS/HEL1OS forecasting problem defined in Section IV, rather than against general-purpose benchmarks.

TABLE I. MODEL FAMILY TRADE-OFFS FOR X-RAY-BASED FLARE FORECASTING

Model Family	Strengths	Limitations in This Context
SVM / Random Forest / XGBoost	Effective on small, structured, tabular feature sets; comparatively interpretable via feature-importance scores.	Do not natively model temporal dependencies across the SXR/HXR time series without extensive manual windowing.
LSTM / BiLSTM	Designed to retain state across time steps, matching the sequential nature of flux time series.	Computationally more expensive to train; internal representations are difficult to interpret without additional analysis.
Transformer-based architectures	Capable of modelling long-range dependencies via self-attention, without the sequential bottleneck of recurrence.	Require substantially larger labelled datasets than are likely to be available from Aditya-L1's current operational history, and are prone to overfitting on rare M/X-class events.
Temporal Convolutional Networks (TCN)	Efficient parallel training; strong performance reported in the literature for short-horizon flux forecasting.	Fixed receptive field can limit sensitivity to multi-scale precursor patterns spanning minutes to hours.

On the basis of this comparison, the LSTM family is selected as the base encoder for both branches in the proposed PG-MBL architecture, primarily because the SoLEXS/HEL1OS precursor signatures described in Section II are expected to unfold over minutes to tens of minutes, a horizon well matched to the gated recurrent state retention that LSTM and related gated architectures were designed to support [2] and because the volume of labelled M/X-class events available from Aditya-L1's operational history to date is unlikely to be sufficient to train a data-hungry Transformer architecture [5] without substantial risk of overfitting. This is a design decision based on the general properties summarized in Table I rather than on a benchmark comparison conducted on Aditya-L1 data, which has not yet been performed.

D. Training Strategy

Flare catalogs are dominated by quiescent intervals, with M- and X-class events representing a small fraction of total observing time. The proposed framework incorporates the following measures to address this imbalance, none of which have yet been empirically validated on Aditya-L1 data:

- Class-weighted or focal loss for the classification head, to prevent the majority quiescent class from dominating the gradient signal during training.
- Stratified temporal splitting for training, validation, and test sets, preserving the chronological order of solar cycle activity so that the model is evaluated on flare events that occur after, not interleaved with, its training period - a precaution against optimistic bias that can arise from randomly shuffled time-series splits.

- Transfer learning from GOES-derived pretraining: given the comparatively short operational history of Aditya-L1 relative to the multi-decade GOES record, the proposed architecture pretrains the SXR branch on GOES soft X-ray data before fine-tuning on SoLEXS, while noting that this requires the two instruments' flux calibrations to be reconciled and that the transferability of features across instruments has not been established for this specific pairing.
- Physics-guided regularization as a soft prior: the Neupert-consistency loss term is weighted such that it acts as a regularizer rather than a hard constraint, allowing the model to deviate from the idealized relationship in cases where the underlying physics itself deviates from it, consistent with the caveat noted in Section II that departures from the Neupert effect are themselves informative rather than purely erroneous [6].

VI. EVALUATION METRICS

Accuracy is not an appropriate primary metric for this problem given the severe class imbalance described in Section V-D; a model that never predicts a flare can achieve high accuracy while providing no operational value. This study therefore presents the metrics summarized in Table II, all of which are standard in the operational space-weather forecasting literature and none of which have yet been computed for the proposed architecture, since it has not been trained or tested.

TABLE II. PROPOSED EVALUATION METRICS

Metric	Definition	Operational Relevance
True Skill Statistic (TSS)	TSS = POD – POFD (Probability of Detection minus Probability of False Detection); ranges from -1 to 1 and is insensitive to class imbalance.	Preferred over accuracy for rare-event forecasting such as X-class flares because it is unaffected by the ratio of flaring to non-flaring intervals.
Heidke Skill Score (HSS)	Measures forecast accuracy relative to a random forecast, normalized against the total number of correct predictions expected by chance.	Enables comparison of the proposed model against climatological or persistence baselines used in prior operational studies.
False Alarm Ratio (FAR)	$FAR = FP / (TP + FP)$, the fraction of predicted flares that did not occur.	Directly tied to operational cost; a high FAR erodes end-user trust in satellite and grid-protection advisories.
Precision	$TP / (TP + FP)$, the fraction of positive predictions that are correct.	Complements FAR and is reported jointly with Recall to characterize the precision–recall trade-off under class imbalance.
Recall (POD)	$TP / (TP + FN)$, the fraction of actual flares correctly identified.	Critical for space-weather safety applications where missed X-class events carry a disproportionately high operational cost.
Lead Time	Interval between the model's alert timestamp and the onset of the	Determines the practical utility of the forecast for satellite safing procedures and astronaut radiation

Metric	Definition	Operational Relevance
	impulsive phase (GOES SXR derivative threshold crossing).	protocols; must be evaluated jointly with FAR to avoid rewarding premature, low-confidence alerts.

Two aspects of this evaluation protocol warrant emphasis. First, the True Skill Statistic and Heidke Skill Score are reported alongside, rather than instead of, Precision and Recall, because TSS and HSS are useful for comparing overall skill against baseline forecasts but can obscure the specific precision–recall trade-off that determines operational usability. Second, Lead Time must be interpreted jointly with False Alarm Ratio: a model can trivially achieve long lead times by issuing frequent, low-confidence alerts well in advance of any flare, at the cost of an unacceptably high false-alarm rate. The proposed evaluation protocol therefore reports a lead-time-versus-FAR curve, analogous to a precision–recall curve, rather than a single lead-time figure, so that the trade-off is visible rather than hidden behind a single summary statistic.

We emphasize that no values for any of these metrics are reported in this study. Any numerical claim of skill would require a trained model evaluated against a held-out, chronologically posterior test set of labelled Aditya-L1 flare events, which is left to future work as described in Section IX.

VII. EXPECTED CONTRIBUTIONS

- A forecasting architecture designed specifically around the thermal/non-thermal separation unique to the SoLEXS/HEL1OS instrument pairing, rather than adapted from magnetogram-centric or single-band X-ray designs.
- An explicit mechanism for embedding the Neupert effect as a trainable physical constraint, intended to reduce the risk of the model learning flux correlations that are physically implausible.
- A feature set : spectral hardness ratio, EM–T trajectory, and SXR derivative, derived specifically from the disk-integrated thermal and non-thermal channels available from Aditya-L1, together with an explicit accounting of the spatial-information limitations this entails.
- A training and evaluation protocol addressing class imbalance and instrument-history scarcity, structured to support a like-for-like comparison against GOES-based baselines once implemented.

These are anticipated contributions of the proposed methodology; none constitute a validated result at this stage.

VIII. LIMITATIONS

- Spatial ignorance: SoLEXS and HEL1OS are disk-integrated instruments and cannot localize a precursor signature to a specific active region. When multiple active regions are present, the model cannot distinguish which region is responsible for an observed precursor pattern, which fundamentally limits the specificity of any resulting forecast.

- Data scarcity for rare events: X-class flares are infrequent by definition, and Aditya-L1's operational history to date is short relative to a full solar cycle, constraining the number of high-magnitude labelled examples available for training and evaluation.
- Unvalidated physical assumptions: the Neupert-effect constraint is drawn from the literature on GOES-era observations; its quantitative form has not yet been characterized for the specific energy bands and cadence of SoLEXS and HELIOS, and the weighting of the physics-guided loss term proposed in Section V-D is a design choice that would require empirical tuning.
- No experimental validation: this study presents an architecture and protocol only. Claims about relative performance of LSTM, tree-based, Transformer, and TCN approaches in Table I are drawn from the general literature surveyed in Section II and have not been independently verified on Aditya-L1 data.

IX. FUTURE WORK

- Construct a labelled Aditya-L1 flare catalog by cross-referencing SoLEXS/HELIOS flux time series against GOES-class labels for the overlapping observation period, to establish a chronologically valid train/validation/test split.
- Train and evaluate the PG-MBL architecture against the metrics in Table II, alongside an ablation study that removes the Neupert-consistency loss term to directly test whether the physics-guided constraint measurably improves TSS, HSS, and FAR relative to an unconstrained multi-branch baseline.
- Conduct a controlled comparison between the proposed two-branch design and a single-branch model trained on concatenated SXR/HXR flux, to empirically test the research-gap hypothesis raised in Section III regarding the value of channel separation.
- Investigate cross-modal fusion with SUI (Solar Ultraviolet Imaging Telescope) data from Aditya-L1 to partially address the spatial-ignorance limitation identified in Section VIII, by using UV bright-point imagery to provide coarse spatial context to the X-ray-derived forecast.
- Evaluate transfer learning from multi-decade GOES soft X-ray archives as a pretraining source for the SoLEXS branch, with explicit cross-calibration of flux units prior to fine-tuning, to mitigate the data-scarcity limitation identified in Section VIII.
- Characterize the EM-T horizontal-branch pattern specifically in SoLEXS-derived spectra, since its behaviour has previously been documented primarily using GOES data and has not yet been verified in this instrument's higher-cadence measurements.

X. CONCLUSION

This study has synthesized the current state of the art in AI-driven solar flare forecasting, distinguishing the well-established magnetogram-centric literature from the comparatively sparse body of work using X-ray flux alone, and has identified a specific research gap concerning the combined thermal/non-thermal X-ray capability unique to Aditya-L1's SoLEXS and HELIOS instruments. This work proposed a Physics-Guided Multi-Branch LSTM architecture that processes the two X-ray channels in parallel branches and constrains their fusion using the Neupert effect [6] as a physically motivated loss term, together with a feature set,

training strategy, and evaluation protocol tailored to the class imbalance and spatial limitations inherent to disk-integrated X-ray forecasting. No experimental results are reported here, and the described architecture, features, and training strategy constitute a proposed methodology rather than a validated system. The future work outlined in Section IX, beginning with the construction of a labelled Aditya-L1 flare catalog, represents the necessary next step toward empirically testing the claims made in this study.

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